3.9 Organic Synthesis

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.9 Organic Synthesis
Difficulty	Easy

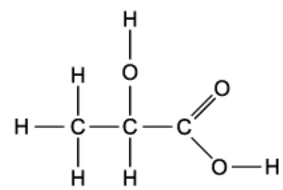
Time allowed: 20

Score: /10

Percentage: /100

Question 1

Lactic acid occurs naturally, for example in sour milk.



Lactic acid

What is a property of lactic acid?

A. It decolourises aqueous bromine rapidly.

B. It is insoluble in water.

C. It reduces Fehling's reagent.

D. It reacts with sodium carbonate.

[1 mark]

Question 2

Which reagent can be used to convert $CH_3CH_2CO_2CH_3$ into $CH_3CH_2CO_2CH_2Cl$?

A. Chlorine in bright sunlight at 100 °C

B. Concentrated hydrochloric acid at 100 °C

C. Phosphorus pentachloride at room temperature

D. Sulphur dichloride oxide (thionyl chloride, $SOCl_2$) at 50 °C

[1 mark]

Question 3

Which reactions produce a coloured organic compound?

1 Ethene + cold dilute acidified potassium manganate (VII)

2 Ethanol + acidified potassium dichromate (VI)

3 Ethanal + 2,4–dinitrophenylhydrazine reagent

A. 1 only

B. 1 and 2

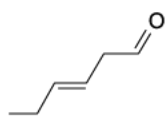
C. 3 only

D. 2 and 3

[1 mark]

Question 4

The characteristic smell of cut grass is due to the compound *trans*-hex-3–enal. The human nose is very sensitive to this compound and is able to detect 0.25 parts per billion in air.



trans-hex-3-enal

Which reagents will react with *trans*-hex-3–enal?

1 Fehling's reagent

2 Sodium borohydride

3 Sodium

A. 1 only

B. 1 and 2

C. 2 only

D. 2 and 3

[1 mark]

Question 5

A compound C has the following properties:

- liquid at room temperature
- soluble in water
- reacts with aqueous sodium hydroxide

What could C be?

A. Propene

B. Propanol

C. Propanoic acid

D. Propyl propanoate

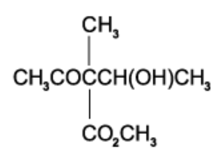
[1 mark]

Question 6

The following compounds are heated under reflux with an excess of hot acidified potassium dichromate (VI).

Which gives a product with a chiral centre?

1



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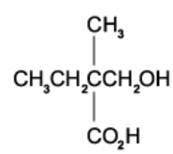

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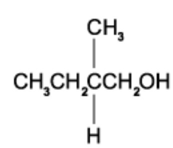


A. 1 only

B. 1 and 2

C. 3 only

D. 2 and 3

[1 mark]

Question 7

Disaccharides, $C_{12}H_{22}O_{11}$, are important in the human diet as they are a source of fast energy. Lactose and sucrose are both disaccharides which can be hydrolysed to release glucose.

$$\text{lactose} + H_2O \rightarrow \underset{\text{glucose}}{CH_2OH(CHOH)_4CHO} + \underset{\text{galactose}}{CH_2OH(CHOH)_4CHO}$$

$$\text{sucrose} + H_2O \rightarrow \underset{\text{glucose}}{CH_2OH(CHOH)_4CHO} + \underset{\text{fructose}}{CH_2OH(CHOH)_3COCH_2OH}$$

Which statements about these hydrolysis products are correct?

1 Both products of lactose hydrolysis are ketones.

2 Both products of lactose hydrolysis are optical isomers.

3 Both products of sucrose hydrolysis have structural isomers.

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 8

Tartaric acid can occur naturally, for example in wine, and may be synthesised in the laboratory.

The laboratory synthesis occurs in 2 steps:

$$\text{OHCCHO} \xrightarrow{\text{step 1}} \text{intermediate} \xrightarrow{\text{step 2}} \text{HO}_2\text{CCH(OH)CH(OH)CO}_2\text{H}$$

tartaric acid

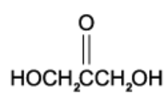
Which reagents could be used for this synthesis?

	Step 1	Step 2
A	HC*l* (aq)	HCN (g)
B	H_2SO_4 (aq)	$K_2Cr_2O_7$ / H_2SO_4 (aq)
C	KCN (aq/alcoholic)	$K_2Cr_2O_7$ / H_2SO_4 (aq)
D	HCN, NaCN (aq/alcoholic)	H_2SO_4 (aq)

[1 mark]

Question 9

DHA is an omega-3 fatty acid which is used to treat heart disease and high cholesterol as well as being essential for brain development during pregnancy.



DHA

Which observations would be made when testing this substance?

1 Hydrogen is produced when sodium is added.

2 A silver precipitate is produced when Tollens' reagent is added.

3 A coloured precipitate is produced when 2,4-dinitrophenylhydrazine reagent is added.

A. 1 only

B. 1 and 3

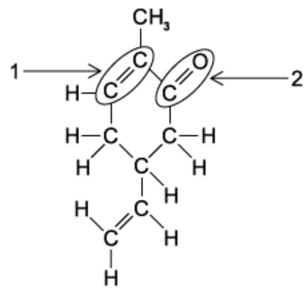
C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 10

This molecule is responsible for the flavour of spearmint chewing gum.



What is a true statement about the functional groups **1** or **2**?

A. **1** will undergo nucleophilic addition

B. **1** will undergo electrophilic substitution

C. **2** will undergo nucleophilic addition

D. **2** will undergo electrophilic substitution

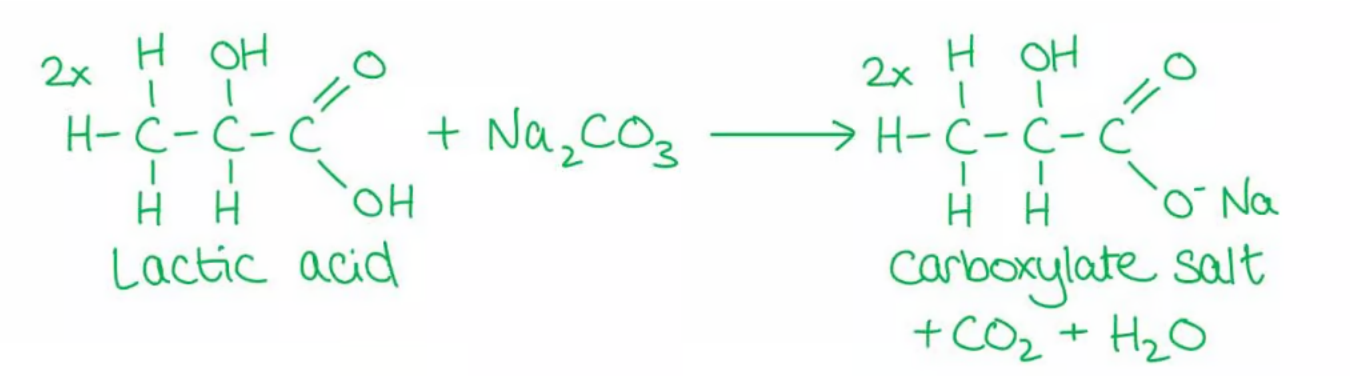
[1 mark]

Model Answers: Easy

1

The correct answer is **D** because:

- Lactic acid contains a **carboxylic acid** (-COOH) group.
- **Sodium carbonate** reacts with carboxylic acids to form a salt, carbon dioxide and water.



A is incorrect as bromine water is **decolorized** when shaken with an **alkene** which undergoes an **addition reaction** with the bromine.

B is incorrect as lactic acid contains two **polar** groups: -COOH (carboxylic acid) and -OH (hydroxyl) groups which can form H-bonds with water. Lactic acid is therefore readily soluble in water.

C is incorrect as only **aldehydes** react with Fehling's reagent by getting oxidised to a **carboxylic acid** and reducing the Fehling's solution. This causes the **clear blue** solution to become an **opaque red/orange** colour.

2

The correct answer is **A** because:

- The **halogenation** of methylpropanoate ($CH_3CH_2CO_2CH_3$) can be achieved by a **free-radical substitution** reaction.

- The hydrogen atom in the alkyl group (-CH_3) is substituted for a halogen.
- The reaction is **initiated** by the formation of two **radicals** using UV light (which comes from the Sun).
  - Radicals are species with unpaired electrons.
- In the **propagation** step the formed radicals attack **reactant molecules** to generate more free radicals.
- In the **termination** step two radicals react together to form an unreactive molecule.

- **Initiation Step:** $Cl_2 \xrightarrow{UV} 2Cl\bullet$
- **Propagation Step:** $CH_3CH_2CO_2CH_3 + Cl\bullet \rightarrow CH_3CH_2CO_2CH_2\bullet + HCl$

$$CH_3CH_2CO_2CH_2\bullet + Cl_2 \rightarrow CH_3CH_2CO_2CH_2Cl + Cl\bullet$$

- **Termination Step:** $2Cl\bullet \rightarrow Cl_2$

$$CH_3CH_2CO_2CH_2\bullet + Cl\bullet \rightarrow CH_3CH_2CO_2CH_2Cl$$

$$2CH_3CH_2CO_2CH_2\bullet \rightarrow CH_3CH_2CO_2CH_2CH_2CO_2CH_2CH_3$$

B, **C** & **D** are incorrect as these reagents are used in the **nucleophilic substitution** reaction of alcohols. $CH_3CH_2CO_2CH_3$ is not an alcohol but an ester.

3

The correct answer is **C** because:

- **Aldehydes** and **ketones** react with 2,4-dinitrophenylhydrazine (2,4-DNPH) in a **condensation** reaction.
  - **Two molecules** join together to form **one molecule** while **eliminating** a small molecule, in this case water.
- This results in the formation of a **deep-orange** precipitate.
- Ethanal is an **aldehyde** and will undergo a condensation reaction with (2,4-DNPH) to form a deep-orange precipitate.

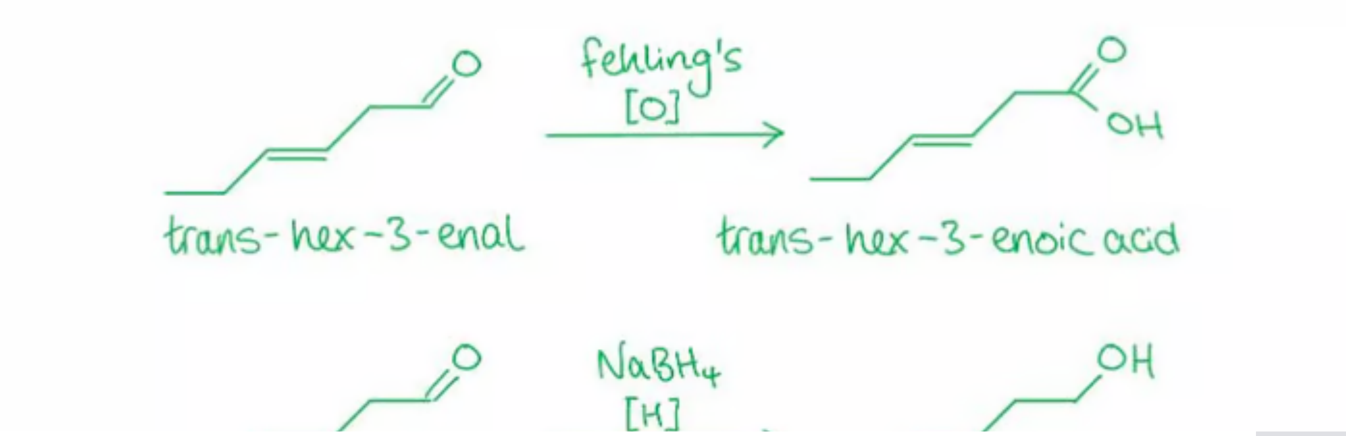
A & **B** are incorrect as alkenes get **oxidised** by cold dilute acidified potassium manganate (VII) this causes the **purple** solution of **Mn^{7+}** ions get reduced to **colourless Mn^{2+}.**

D is incorrect as acidified potassium dichromate (VI) oxidises primary alcohols, secondary alcohols and aldehydes. The orange solution containing dichromate (VI) ions are reduced to green solution containing chromium (III) ions. Chromium ions are not an **organic compound.**

4

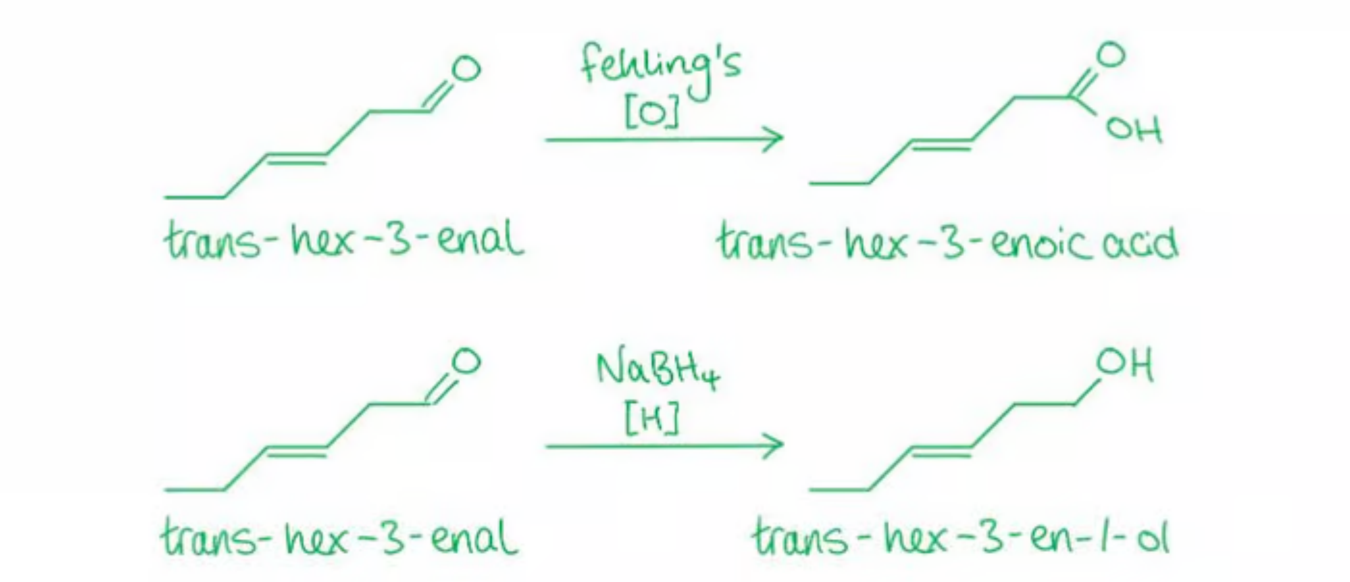
The correct answer is **B** because:

- The Fehling's reagent is an **alkaline** solution containing copper (II) ions and is an oxidising agent.
  - An oxidising agent is a species that oxidises another species while itself gets reduced.
  - When warmed with an **aldehyde**, the Cu^{2+} are reduced to Cu^{+} changing the clear **blue** Fehling's solution colour to an opaque red/orange colour as the precipitate of copper (I) oxide forms.
- Sodium borohydride ($NaBH_4$) is a reducing agent which reduces **carboxylic acids, aldehydes** and **ketones.**
  - A reducing agent is a species that reduces another species while itself gets oxidised.
- The aldehyde in trans-hex-3-enal gets oxidised by Fehling's reagent to a **carboxylic acid.**
- The aldehyde in trans-hex-3-enal gets reduced by $NaBH_4$ to a **primary alcohol.**


fehling's
[O]
trans-hex-3-enal
trans-hex-3-enoic acid
NaBH4
[H]
OH

acid.

- The aldehyde in trans-hex-3-enal gets reduced by $NaBH_4$ to a **primary alcohol.**



A & **C** are incorrect as trans-hex-3-enal reacts with two of the three reagents.

D is incorrect as trans-hex-3-enal doesn't contain an **alcohol** or **carboxylic acid** to react with the reactive metal and form a **salt** and **hydrogen.**

5

The correct answer is **C** because:

- Propanoic acid is a **liquid** at room temperature.
- The carboxylic acid group (-COOH) is highly polar and forms H-bonds with water, causing propanoic acid to be soluble in water.
- Propanoic acid is an **acid** which will react with aqueous sodium hydroxide, which is a **base**, in an acid-base reaction.

A is incorrect as **propene** is a **gas** at room temperature. It is **insoluble** in water as it is non-polar. It doesn't react with aqueous sodium hydroxide.

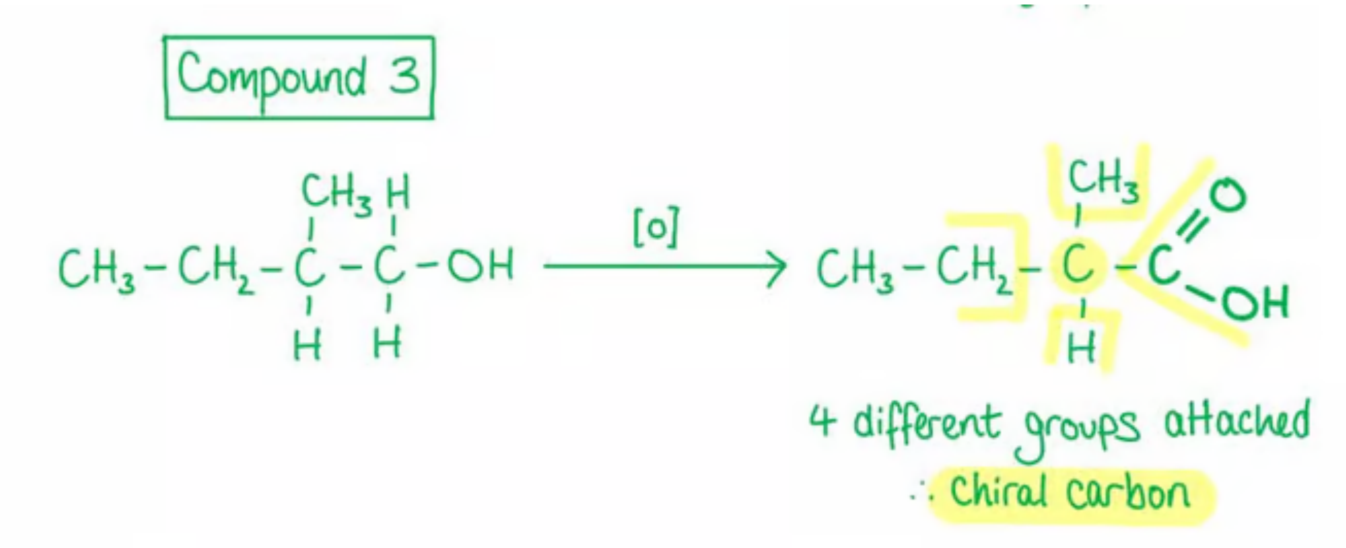
B is incorrect as the hydroxide ions in sodium hydroxide is a strong base which will react with an acid. P**ropanol** is a base and will therefore not react with sodium hydroxide.

D is incorrect as the two oxygen atoms can form H-bonds with water, however the big non-polar alkyl groups make it **insoluble** in water.

6

The correct answer is **C** because:

- Hot acidified potassium dichromate(VI) **oxidises** primary alcohols, secondary alcohols and aldehydes to carboxylic acids, ketones and carboxylic acids respectively.
- The primary alcohol in the third compound will get oxidised to a carboxylic acid.
- The product will contain a chiral centre.
  - A chiral centre is a carbon with four different atoms or groups of atoms.



A is incorrect as compound 1 contains a secondary alcohol which is oxidised to a **ketone.** The product only has 3 different atoms or groups of atoms (two ketone groups) and has therefore not a chiral centre.

B & **D** are incorrect as compound 2 contains a primary alcohol which is oxidised to a **carboxylic acid.** The product only has 3 different atoms or groups of atoms (two carboxylic acids) and has therefore not a chiral centre.

7

The correct answer is **C** because:

- Glucose and galactose contain four carbon atoms with four different atoms or groups of atoms.

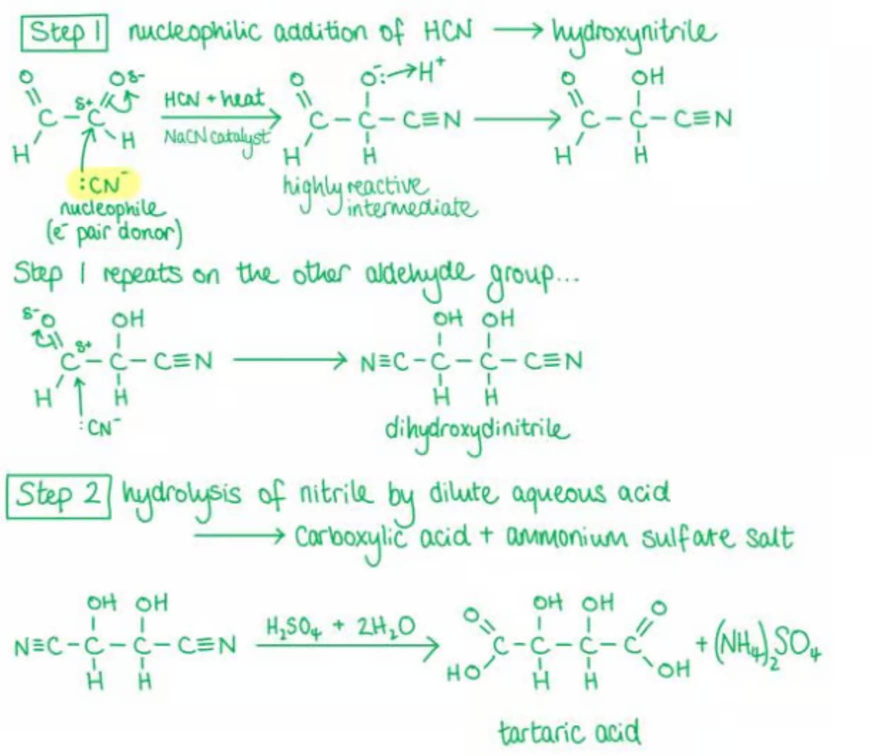
- They therefore have four **chiral centres** and can exist as **optical isomers.**
    - **Optical isomers** are stereoisomers that can rotate the plane of polarized light.
- **Structural isomers** are compounds with the same molecular formula but different structural formulae.
- Both glucose and fructose can exist in the cyclic form.

A, B & **D** are incorrect as both glucose and galactose are **aldehydes.**

8

The correct answer is **D** because:

- The first step involves a nucleophilic addition of CN^- using NaCN as catalyst and heat to form a **hydroxynitrile.**
- In the second step, the nitrile is **refluxed** with dilute aqueous sulfuric acid causing hydrolysis of the nitrile forming a carboxylic acid and ammonium salt.

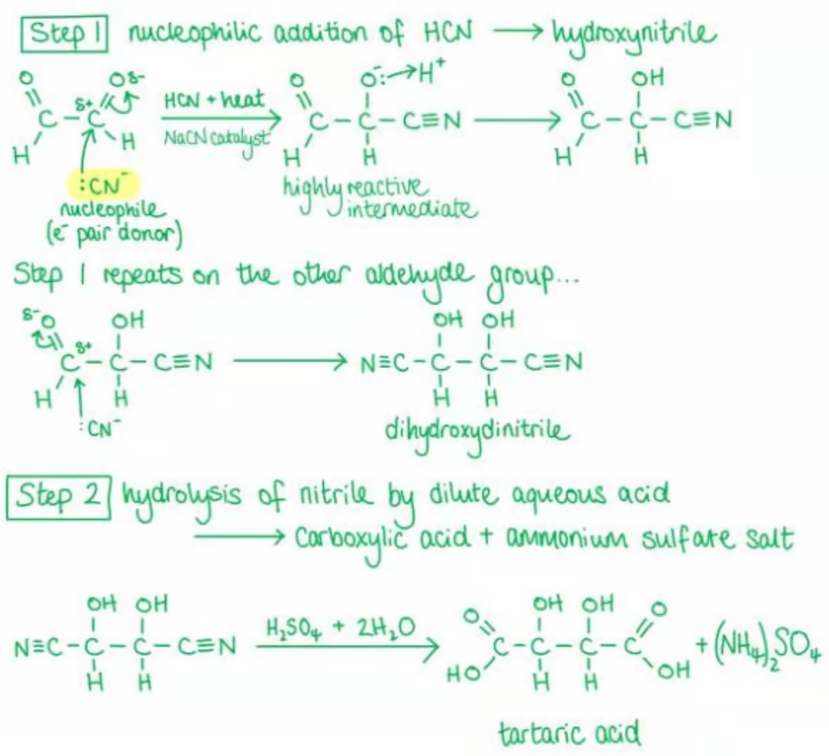


A is incorrect as **nucleophilic addition** of Cl^- followed by **nucleophilic substitution** by CN^- gives a hydroxynitrile.

B is incorrect as these are the reverse steps for the **oxidation** of aldehydes to **carboxylic acids.**

C is incorrect as **nucleophilic addition** of CN^- takes place to form a hydroxynitrile. The secondary alcohol in the hydroxynitrile is then oxidised to a ketone by $K_2Cr_2O_7$ / H_2SO_4 (aq).

9



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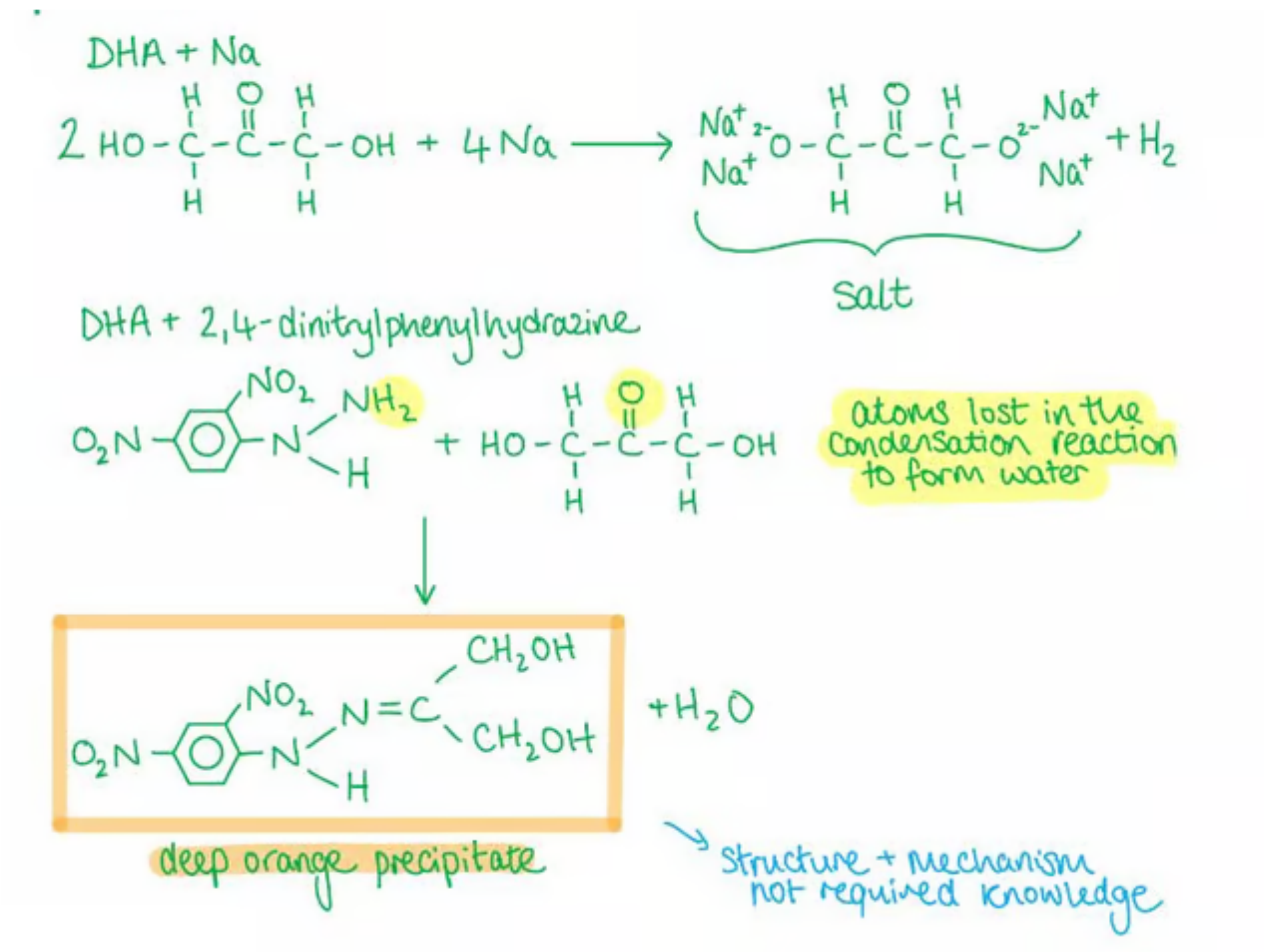
9

The correct answer is **B** because:

- **Na** reacts with alcohols and carboxylic acids forming a salt and hydrogen gas.
  - **DHA** has two alcohol groups which both will react with the sodium metal.
- **2,4-dinitrophenylhydrazine** identifies carbonyl compounds by undergoing a **condensation reaction** when heated with the compound and forming a **deep-orange** precipitate and water.



A is incorrect as **DHA** also reacts with 2,4–dinitrophenylhydrazine.

C & D are incorrect as Tollens' reagent tests for the presence of **aldehydes** by oxidising it to a **carboxylic acid.** The Ag^+ ions themselves are reduced to Ag atoms forming a silver 'mirror' on the inside of the tube. DHA does not contain any aldehyde groups.

10

The correct answer is **C** because:

- The oxygen is more electronegative and therefore has a stronger pull on the electrons in the C=O bond.
- This leaves the carbon slightly positively charged and the oxygen slightly negatively charged.
- The carbonyl carbon is attracted to **nucleophiles** and will undergo **nucleophilic addition.**

- A nucleophile attacks the carbon atom in the carbonyl group and addition across the C=O bond occurs.

A is incorrect as the double bond in C=C in group **1** is rich in electrons and will therefore be attracted to **electrophiles.** Group **1** will undergo **electrophilic addition.**

B is incorrect as though group **1** will be attracted to **electrophiles** it will not undergo **substitution** reactions as there are no groups that can be replaced. It will undergo **addition** reactions to the C=C bond.